Question Paper

Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SEVENTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION
22nd-November- 2024

Error Control Coding [ECE 4073]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) Construct a group G under modulo-11 multiplication and find whether 2 and 4 are generators of the constructed group. (4)

A)

B) Let **a** be a nonzero element in a finite field GF(q). Let ***n*** be the order of **a**. Then, n divides q – 1. Prove this proposition with an example. (3)

C) The minimal polynomial m(X) of an element β in GF(2^m) is irreducible over GF(2). Prove this with an example. (3)

2) The parity check equations for a systematic LBC are given by (4)

A) v0 = u0 + u1 + u3,
v1 = u0 + u2 + u3,
v2 = u0 + u1 + u2,
v3 = u1 + u2 + u3.
Determine how many error patterns it can correct? Determine the syndrome if $r(x)=1+x^8+x^6$. If error is detected, estimate the error pattern and the message vector.

B) For a (7,4) d=3 construction, show that Singleton bound and Gilbert-Varshmov bounds are satisfied. (3)

C) Construct parity equations to encode 6 bit message using the non-systematic Hamming code. Hence determine the codeword for the message 110011. Correct the received vector (1010110101) for single bit error if any. (3)

3) Consider the digital transmission over a frequency flat fading channel. The channel coherence time is ***5ms.*** The coded information bits are transmitted at a rate of 25 kbps. (4)

A) a) The system adopts a linear binary block code with codeword length 7 that is capable of correcting single errors.
b) Determine the minimum interleaver depth and the maximum decoding delay if we want each codeword to contain at most one-bit error from the same error burst. What if the code can correct double errors, design the depth of the interleaver.

B) Consider a cyclic code with (7,4), with $\mathbf{g(x)=1+x+x^3}$, obtain the encoded sequence for m=0011, also explain the decoding procedure. (3)

C) Explain the general structure of a Meggitt decoder using any (7,4) cyclic code. (3)

4) Consider a (15,7,5) Double error correcting BCH code given that $\mathbf{g(x)=x^8+x^7+x^6+x^4+1,}$ if $\mathbf{m(x)=1+x}$ and at the receiver C=[000001101111011]. Detect and correct the two errors using PGZ algorithm. (5)

A)

B) Consider a triple error correcting RS code over GF(2^4), $p(x)=1+x+x^4$, the syndrome computed for (3)

the received polynomial are $\{\beta^{12},1,\beta^{14},\beta^{10},0,\beta^{12}\}$. Determine the error locator polynomial.

C) Design a RS code capable of correcting 10 symbol errors. Choose GF(2^8) as the field and find d_{min}. (2)

5) Consider a message sequence 10011 with **g_1=(1 1 1) and g_2=(1 0 1)**. Find the codeword using (5)

A) Time domain and Transform domain approaches. Write the block diagram of the encoder and mark **(n,k,l).**

B) Consider X1=M0+M1+M2 and X2=M0+M2. Design a trellis encoder, write the state diagram. (3)

C) From the above state table obtained from (Q5B), if the received bits at the decoder are 0110110000, correct the erroneous bit using Viterbi decoding algorithm and trace the path. (2)

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